E.G.S.PILLAYARTS&SCIENCECOLLEGE(AUTONOMOUS)
NAGAPATTINAM–611002.TAMILNADU
(Accredited by NAAC With "A" Grade)
(MBA–ApprovedbyAICTE,NewDelhi)
(AffiliatedtoBharathidasanUniversity,Tiruchirappalli–24)

DEPARTMENT OF BIOTECHNOLOGY

Choice Based Credit System – Learning Outcomes Based Curriculum Framework (CBCS – LOCF)

(Applicable to the candidate admitted from the academic year 2026 – 2027 onwards)

EDAYATHANGUDY G.S. PILLAY ARTS & SCIENCE COLLEGE
(An Autonomous Institution, Affiliated to Bharathidasan University, Tiruchirappalli) Nagore Post, Nagapattinam – 611 002, Tamil Nadu
M.Sc. BIOTECHNOLOGY CURRICULUM STRUCTURE
(Applicable to the candidates admitted from the academic year 2025-26 onwards)

Sem.	CourseS	Title	Ins. Hrs/ Week	Credits	Exam. Hrs.	Marks		Total
						Int.	Ext	
I	Core Course-I (CC)	Cell and Molecular Biology	6	5	3	25	75	100
	Core Course-II(CC)	Biochemistry	6	5	3	25	75	100
	Core Course-III(CC)	Microbiology	6	5	3	25	75	100
	Core Practical – I (CP)	Cell and Molecular Biology, Biochemistry, Microbiology(P)	6	4	3	50	50	100
	Elective Course-I(EC)	A. Enzymology B. Developmental Biology	6	4	3	25	75	100
	Total		**30**	**23**	-	-	-	**500**
II	Core Course-IV(CC)	Immunology	6	5	3	25	75	100
	Core Course-V(CC)	Genetic Engineering	6	5	3	25	75	100
	Core Course -VI(CC)	Plant and Animal Biotechnology	6	5	3	25	75	100
	Core Practical – II (CP)	Immunology, Genetic Engineering, Plant and Animal Biotechnology (P)	6	4	6	50	50	100
	Elective Course-II(EC)	A. Medical Biotechnology B. Nano Biotechnology	4	4	3	25	75	100
	Non-Major Elective Course-I	A. Molecular Biotechnology B. Healthcare Technology	2	2	3	25	75	100
	Total		**30**	**25**	-	-	-	**600**
III	Core course-VII(CC)	Research Methodology	6	5	3	25	75	100
	Core course-VIII(CC)	Bioprocess Technology	6	5	3	25	75	100
	Core course -IX(CC)	Environmental Biotechnology	6	5	3	25	75	100
	Core Practical – III (CP)	Bioprocess Technology, Environmental Biotechnology(P)	6	4	6	50	50	100
	Elective Course III(EC)	A. Food Biotechnology B. Microbial biotechnology	4	4	3	25	75	100
	Non-Major Elective Course-II	A. Agricultural Biotechnology B. Marine Biotechnology	2	2	3	25	75	100
	Internship	Internship	-	2*	-	20	80	100
	Total		**30**	**25**	-	-	-	**700**
IV	Core Course-X(CC)	Bioinformatics and Bioethics	6	5	3	25	75	100
	Core Course-XI(CC) -	Bioinstrumentation	6	5	3	25	75	100
	Skill Oriented Course I (SOC)	Bioenterpernership, Biopatenting, Biosafety	6	5	3	25	75	100
	Project	Project	12	5	-	20	80	100
	Total		**30**	**20**	-	-	-	**400**
Grand Total			**120**	**93**	-	-	-	**2200**

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SUMMARY OF CURRICULUM STRUCTURE OF PG PROGRAMMES

SL.NO	TYPES OF THE COURSES	NO OF COURSES	NO.OF CREDITS	MARKS
1.	Core Courses	11	55	1100
2.	Core Practicals	3	12	300
3.	Elective Courses	3	12	300
4.	Internship	1	2*	100
5.	Project	1	5	100
6.	Non Major Elective Courses	2	4	200
7.	Skill Oriented Courses	1	5	100
Total		**22**	**93**	**2200**

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SEMESTER I

First Year	CORE COURSE –I (CC)	Credit : 5
Semester - I	CELL AND MOLECULAR BIOLOGY	Sub Code: P25BTCC11

(Theory)

LEARNING OBJECTIVES

- Familiarize the student in different areas of cell and molecular biology streams ,
- including cellular organization and their interactions.
- Study the DNA replication, Protein biosynthesis and translational regulation.
- Develop a comprehensive understanding of the complete cellular and molecular
- function of cell organelles in cell – to – cell interaction, gene regulation, and cellular signaling.

UNIT	CONTENTS	HOURS
I	Chemical Composition of Membranes; Membrane Lipids, Membrane Carbohydrates; Structure and Functions of Membrane Proteins; Integral Membrane Proteins, Peripheral Membrane Proteins, Membrane Fluidity; Importance of Membrane Fluidity, Maintaining Membrane Fluidity, Lipids Rafts.Movement of Substances across Cell Membranes; the Energetic of Solute Movement, Diffusion of Substances through Membranes, Facilitated Diffusion, Active Transport..	10
II	Mitochondria : structure, origin and evolution, organization of respiratory of respiratory chain complexes, Mitochondrial Genome; Structure – functional relationship; Structure and function of peroxisome; Structure and function of microbodies, Golgi apparatus, Lysosomes and Endoplasmic reticulum; Organization and role of microtubules and microfilaments; Intermediate filaments; Cellular motility; Molecular motors. Nucleus: Structure and function of nuclear envelope, lamina and nucleolus; Chromatin organization and packaging.	15
III	Models of DNA replication: Semiconservative DNA replication, semidis continuous DNA replication, unidirectional and bidirectional DNA replication – Mechanism of DNA replication in Prokaryotes, DNA replication in Eukaryotic cells. DNA Repair – Nucleotide excision repair, Base excision repair, Mismatch repair, Double strand Breakage Repair. Recombination: double strand break model or recombination.	15
IV	An overview of the flow of Information through the cell – Overview of Transcription: Transcription in Bacteria, Transcription and RNA processing in Eukaryotic cells – Synthesis and processing of Ribosomal and Transfer RNA,	10

	Small noncoding RNAs and RNA interference. Translating Genetic information: Initiation, Elongation, Termination.	
V	Basic properties of a Cancer cell – Causes of cancer – Genetics of cancer – Tumor suppressor Genes and Oncogenes: Tumor viruses and oncogenes: Transformed cells, detection of integral viral DNA, structure of integral viral DNA. Protein kinase and transformation by retro viruses.Cellular counterpart of Carcinogens.Activation of oncogenes.Oncogenic proteins – protein kinase, growth factors, ras protein. Cancer Genome, New strategies for combating cancer – Immunotherapy, Inhibiting the activity of cancer promoting proteins.	**10**

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts of cell and molecular biology and their significance in living systems.	**K1**
CO2	Describe the structure and functions of prokaryotic and eukaryotic cells and their organelles	**K2**
CO3	Explain the structure, replication and organization of DNA and RNA.	**K2**
CO4	Analyze the mechanisms of gene expression including transcription and translation.	**K3**
CO5	Apply the principles of cell division and molecular biology techniques in biological research.	**K4**
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Freifelder D, 2004, Molecular Biology, Narosa Publishing House, New Delhi.
- Lewin B. 2018. Genes XII. Oxford University Press, London.
- Ajoy Paul. 2014. Textbook of Cell and Molecular Biology. Books and Allied Ltd.
- Karp, G., Iwasa, J., & Marshall, W, (2018). Karp's Cell Biology. John Wiley & Sons.
- Alberts, B., Bray, D., Hopkin, K., Johnson, A. D., Lewis, J., Raff, M., & Walter, P. (2019).
- Lodish, H., Berk, A., Kaiser, C.A., Kaiser, C., Krieger, M., Scott,M.P.0&Matsudaria, P. (2008). Molecular cell biology. Macmillan.
- Thomas D. Pollard & William C. Earnshaw& Graham T. Johnson. 2016. Cell Biology. 3rd Edition. Elsevier Health Sciences.
- Watson JD, Gilman M, Witkowski J and Zoller M. 2010. Recombinant DNA. Scientific American Books. 4th Edition. New York.
- Blackburn GM and Gait MJ. 2022. Nucleic Acids in Chemistry and Biology 4th edition. Oxford University Press.

- Lodish H, Baltimore D, Beck A, ZipurskySl, Matsudaria P and Darnell J. 2007. Molecular Cell Biology. 4th Edition. Scientific American Books.
- CooperM 2007. The Cell Molecular Approacj. 2nd Edition. ASM Press.
- Lewis J Kleinsmith and Valerie M Kish. 1995. Principle of Cell and Molecular Biology 2nd Edition. Benjamin – Cummings Publishing Company.
- De Robertis, EDP and E.M>F Roberties. 1987. Cell and Molecular Biology. 8th Edition. Saunders Company.

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First Year	CORE COURSE – II (CC)	Credit : 5
Semester - I	BIOCHEMISTRY	Sub Code: P25BTCC12

(Theory)

LEARNING OBJECTIVES

- Familiarize the student in different areas of cell and molecular biology streams ,
- including cellular organization and their interactions.
- Study the DNA replication, Protein biosynthesis and translational regulation.
- Develop a comprehensive understanding of the complete cellular and molecular
- function of cell organelles in cell – to – cell interaction, gene regulation, and cellular signaling.

UNIT	CONTENTS	HOURS
I	pH, PK . acid, base .Buffers- Henderson- Haselbach equation, biological buffer system Phosphate buffer system, protein buffer system, bicarbonate buffer system, amino acid buffer system and Hb buffer system. Water, Carbohydrates: Nomenclature, classification, structure,chemical and physical properties of carbohydrates. Metabolisms: glycogenesis, glycogenolysis, gluconeogenesis, pentose phosphate pathway	10
II	Lipids: Nomenclature, classification, structure, chemical and physical properties of fatty acids. Metabolisms: biosynthesis of fatty acids, triglycerols. phospholipids, glycol lipids. Cholesterol biosynthesis, bile acids and salt formation. Eicosanoids, sphingolipids and steroid hormones.	15
III	Bioenergetics — Concept of energy, Principle of thermodynamics, Relationship between standard free energy and Equilibrium constant, ATP ads universal unit of free energy in Biological systems. Biological oxidation: Electron transport chain, oxidative phosphorylation, glycolysis, citric acid cycle, cori.s cycle, glyoxalate pathway. Oxidation of fatty acids- mitochondrial and peroxisomal ß-oxidation, alpha and beta oxidation, oxidation of unsaturated and odd chain fatty acids, ketone bodies. Photosynthesis, urea cycle, hormonal regulation of fatty acids and carbohydrates metabolisms, Mineral metabolism.	15
IV	Amino acids and Protein: Nomenclature, Classification, structure, chemical and	10

	physical properties of amino acids and proteins. Metabolisms: Biosynthesis of amino acids. Degradation of proteins, nitrogen metabolisms and carbon skeleton of amino acids. Over all in born error metabolisms.	
V	Nucleic acids: Nomenclature, Classification, structure, chemical and physical properties of purine and pyrimidines. In de novo and salvage synthesis of purines, pyrimidine bases, nucleosides and nucleotides. Catabolisms of purines and pyrimidines bases. Synthetic analogues of nitrogenous bases.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts of biochemistry and the chemical composition of living organisms.	K1
CO2	Describe the structure and functions of carbohydrates, proteins, lipids and nucleic acids.	K2
CO3	Explain the role of enzymes, vitamins and hormones in biological systems.	K2
CO4	Analyze the metabolic pathways of carbohydrates, proteins and lipids.	K3
CO5	Apply biochemical principles and laboratory techniques for the study of biomolecules and metabolism.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Philip Kuchel, Simon Easterbrook-Smith, Vanessa Gysbers, Jacqui M. Matthews, 2011. Schaum.s Outline of Biochemistry, Third Edition (Schaum.s Outline Series), McGraw-Hill.
- Sathyanarayana.U and U.Chakrapani., 2011. Biochemistry. Books and Allied private limited, Kolkata.
- Jeremy M. Berg, John L. Tymoczko, Lubert Stryer, 2010. Biochemistry, Seventh Edition, W. H. Freeman.
- Albert Lehninger, David L. NelsonVoet Donald, Judith G.Voet and Charlotte W.Pratt., 2008. Principles of Biochemistry. John Wiley and sons, Inc., New Jersey.
- Michael M. Cox, 2008. Lehninger Principles of Biochemistry, Fifth Edition, W. H. Freeman publishers.

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First Year	CORE COURSE – III (CC)	Credit : 5
Semester - I	MICROBIOLOGY	Sub Code: P25BTCC13

(Theory)

LEARNING OBJECTIVES

- To understand the structure, functions different types of microscope.
- To identify and Safety guidelines in Microbiology Laboratories.
- To understand distribution, cultivation and life cycle of algae .
- Describe the bacterial growth of classification and identification.
- Explain the concept and applications of microbial diversity.

UNIT	CONTENTS	HOURS
I	History and Scope of Microbiology. Microscopy– Principles and applications. Types of Microscopes - Bright field, Dark- field, Phase-contrast, Fluorescence microscope, Transmission electron microscope (TEM) and Scanning electron microscope (SEM). Sample preparation for SEM & TEM. Atomic force, Confocal microscope. Micrometry – Stage, Ocular and its applications .	10
II	Safety guidelines in Microbiology Laboratories. Sterilization, Disinfection and its validation. Staining methods – Simple, Differential and Special staining. Automated Microbial identification systems - Pure cultures techniques – Cultivation of Anaerobic organisms. Maintenance and preservation of pure cultures. Culture collection centres - National and International.	15
III	Distribution, morphology, classification, reproduction and economic importance. Isolation of algae from soil and water. Media and methods used for culturing algae, Strain selection and large-scale cultivation. Life cycle - Chlamydomonas, VolvoxSpirogyra (Green algae), N (Red algae	15
IV	Bacterial Structure, properties and biosynthesis of cellular components – Cell wall. Actinomycetes and Fungi - Distribution, morphology, classification, reproduction and economic importance. Sporulation. Growth and nutrition - Nutritional requirements, Growth curve, Kinetics of growth, Batch culture, Synchronous growth, Measurement of growth and factors affecting growth.	10
V	Introduction to microbial biodiversity – Thermophiles - Classification, Thermophilic Archaebacteria and its applications. Methanogens - Classification, Habitats, applications. Alkaliphiles and Acidophiles - Classification, discovery basin, its cell wall and membrane. Barophiles - Classification and its applications. Halophiles - Classification, discovery basin, cell walls and membranes – purple membrane, compatible solutes. Microbial stress response –Osmoadaptation / halotolerance - Applications of halophiles.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the science of microbiology, its scope and importance in human welfare	K1
CO2	Describe the history, classification and characteristics of microorganisms.	K2
CO3	Explain the structure, growth and reproduction of microorganisms	K2
CO4	Analyze the factors affecting microbial growth and microbial interactions	K3
CO5	Apply aseptic techniques, sterilization methods and microbiological laboratory practices.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Kanunga R. (2017). Ananthanarayanan and Panicker's Text book of Microbiology. (10th Edition). Universities Press (India) Pvt. Ltd.
- Chan E.C.S., Pelczar M. J. Jr. and Krieg N. R. (2010). Microbiology. (5th Edition). Mc.Graw Hill. Inc, New York
- Prescott L. M., Harley J. P. and Klein D. A. (2004). Microbiology. (6th Edition). McGraw Hill company, New York.
- White D. Drummond J. and Fuqua C. (2011). The Physiology and Biochemistry of Prokaryotes, Oxford University Press, Oxford, New York.
- Dubey R.C. and Maheshwari D. K. (2009). Textbook of Microbiology. S. Chand, Limit
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- Webster J. and Weber R.W.S. (2007). Introduction to Fungi. (3rd Edition). Cambridge University Press, Cambridge.
- Schaechter M. and Leaderberg J. (2004). The Desk encyclopedia of Microbiology. Elseiver Academic Press, California.

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First Year	CORE PRACTICAL – I (CP)	Credit : 4
Semester - I	CELL AND MOLECULAR BIOLOGY, BIOCHEMISTRY, MICROBIOLOGY (P)	Sub Code:P25BTCP1

(Practical)

Objective:

- Study the Physiology of the cells and their enumeration.
- Identify the specialized chromosomes.
- Perform the isolation, purification and quantification of nucleic acids

PRACTICALS

CELL AND MOLECULAR BIOLOGY

1. Cell Division – Mitosis and Meiosis
2. Barr body Identification in cells of Buccal smear.
3. Isolation and Purification of Genomic DNA from prokaryotes.
4. Separation of DNA by AGE.
5. Separation of protein by SDS- PAGE.
6. Conjugation, Transformation and Transduction (Demo)

BIOCHEMISTRY

1. Prepration of Acetate and phosphate buffer system and validate the HendersonHasselbach equation.
2. To determine concentration of an unknown protein by UV-Vis Spectrophotometer.
3. Determination of Molar extinction coefficient of protein.
4. Protein purification by gel filtration, ion-exchange chromatography
5. Determine pKa and pI of amino acids
6. Protein separation by SDS-PAGE
7. Qualitative and Quantitative Analysis of Carbohydrates, Amino acids and proteins, lipid and nucleic acid.
8. Separation of amino acids and sugars by TLC
9. Extraction of proteins, RNA and DNA from cultured cells.

MICROBIOLOGY

1. Isolation and identification of bacteria :
   a. Cultural characteristics of bacteria on NA.
   b. Pure culture techniques (Types of streaking).
   c. Staining techniques: Gram's, Negative, Endospore, Capsule and Cell Wall.
2. Biochemical characterization of bacteria : IMViC, Carbohydrate fermentation test, Mannitol motility test, Gelatin liquefaction test, Urease test, TSI test, Nitrate reduction test, Catalase test, Oxidase test, Starch hydrolysis, Casein hydrolysis.
3. Isolation and characterization of acidophilic, alkalophilic and halophilic bacteria.
4. Identification of fungi by lactophenol cotton blue method.
5. Production and quantification of biofilm by using microorganisms.

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Identify and use common laboratory instruments, equipment and safety procedures in biological science laboratories.	K1
CO2	Describe the principles and procedures involved in microbiological, biochemical and cell biology experiments	K2
CO3	Perform staining, culturing, biomolecule analysis and molecular biology techniques using standard laboratory protocols	K3
CO4	Analyze experimental results related to microorganisms, biomolecules and cellular components.	K3
CO5	Apply practical skills for the isolation, identification and characterization of biological samples using appropriate laboratory techniques.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References:

1. James G. Cappuccino and Natalie Sherman. Microbiology: A Laboratory Manual.
2. Kanika Sharma. Manual of Microbiology: Tools and Techniques.
3. Robert S. Burlage, Ronald Atlas, David Stahl, Gill Geesey, Gary Sayler, 1998. Techniques in Microbial Ecology, Oxford University Press. N.Y.
4. Samuel Singer, 2001. Experiments in Applied Microbiology, Academic Press

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First Year	ELECTIVE COURSE – I (EC)	Credit : 4
Semester - I	A.ENZYMOLOGY	Sub Code:P25BTEC1A

(Theory)

LEARNING OBJECTIVES

- To understand the structure, function sand mechanism of action of enzymes
- To identify and Classify enzymes based on the reactions catalysed.
- Understand kinetics, inhibition and regulation of enzyme catalysed reactions.
- Describe the concept of Regulation of Enzyme activity.
- Explain the concept and appplications of enzymes.

UNIT	CONTENTS	HOURS
I	**Introduction to enzymes**:Holoenzyme, apoenzyme, and prosthetic group; Interaction between enzyme and substrate- lock and key model, induced fit model. Features of active site, activation energy, Rate enhancement through transition state stabilization, Chemical mechanism for transition state stabilization. Enzyme specificity and types. Enzyme Commission system of classification and nomenclature of enzymes (Class subclass and sub sub class with one example). Ribozymes,Abzymes.	10
II	**Enzyme kinetics:** Importance, order of reaction, studyof the factors affecting the velocityof enzyme catalyzed reaction-enzyme concentration, temperature, pH, substrate concentration, inhibitors and activators (explanation with graphical representation). Derivation of Michaelis Menten equation and Km value and its significance, Definition of V_{max}value of enzyme and its significance, Lineweaver-Burk plot; King and Altman Method to determine velocity Equations.	15
III	**Enzyme inhibition:** Reversible and irreversible-examples. Reversible-competitive, non-competitive and uncompetitive inhibition; Graphic determination of inhibitor type, Dose- response curves of Enzyme inhibition. Mutually Exclusive Binding of Two inhibitors; Structure-Activity Relationships and Inhibitor Design.	15
IV	**Regulation of Enzyme activity:** Covalently modulated enzymes with examples of adenylation and phosphorylation; Zymogen form of enzyme and zymogen activation; Multienzyme complexes and their role in regulation of metabolic pathways; Allosteric regulation - Aspartate trascarbamoylase. Isoenzymes- Lactate dehydrogenase and creatine phosphokinase.	10
V	**Application of enzymes:** Industrial uses of enzymes: production of glucose from starch, celluloseand dextrans, use of lactase in dairy industry, production of glucose fructose syrup fromsucrose, use of proteases in food, leather and detergent industry. Diagnostic and therapeutic enzymes; Enzyme engineering	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts, classification and nomenclature of enzymes.	K1
CO2	Describe the structure, properties and mechanism of enzyme action.	K2
CO3	Explain enzyme kinetics, factors affecting enzyme activity and enzyme inhibition.	K3
CO4	Analyze the role of enzymes in metabolic pathways and biological processes.	K3
CO5	Apply enzymological principles and techniques for enzyme assay and industrial applications.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- FundamentalsofEnzymology:ThecellandmolecularBiologyofCatalyticProteinsbyNicholas C. Price, Lewis Stevens, and Lewis stevens, Oxford University Press, USA
- Enzyme Kinetics: A modern Approach, AlejandroG. Marangoni,Wiley-Interscience
- Enzyme Mechanism by P.KSivaraj Kumar, RBSA Publishers
- Enzymes in Industry: Production And Applications by Aehle W(2007) John Wiley & Sons Inc
- Enzymes:Biotechnology,ClinicalChemistry(secondEdition)byTrevorPalmer,Philip Bonner Horwood Publishing Limited
- Lehninger, Principles of Biochemistry DavidL.Nelson Michael M.Cox, W.H. Freeman.
- Biochemistry DonaldVoet, Judith G.Voet Publisher: JohnWiley&SonsInc
- Enzymology Dixonand Webb, Academic Inc publishers, Newyork.

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First Year	ELECTIVE COURSE – I (EC)	Credit : 4
Semester - I	DEVELOPMENTAL BIOLOGY	Sub Code:P25BTEC1B

(Theory)

LEARNING OBJECTIVES

- To provide the basic process of early embryonic development.
- To emphasize embryonic organizers, the nature of genetic information during cell differentiation
- To highlight gametogenesis and its regulation.
- To compare the types of eggs, cleavage patterns, and developmental fate.
- To discuss the molecular regulation of organogenesis and regeneration in amphibians and reptiles.
- To relate the role of hormones, the aging process and genetic disorders disrupting normal development.

UNIT	CONTENTS	HOURS
I	Introduction - potency - commitment - specification - induction competence - determination -differentiation. Morphogenetic gradients - ce fate and cell lineages. Stem cells. Genomic equivalence and cytoplasm determinants. Imprinting. Mutants and transgenics in analysis development - organizers - classical experiments on organizers.	10
II	Gametogenesis – spermatogenesis – spermiogenesis - structure and types of sperm. Oogenesis – origin and growth of oocyte - maturation of egg - vitellogenesis - types of eggs. Polarity and symmetry of an egg - organization of egg cytoplasm. Fertilization - events of fertilization - acrosome reaction in sperm – egg activation during fertilization - theories of fertilization.	15
III	Cleavage - patterns of cleavage -blastula formation - embryonic fields - gastrulation - morphogenetic movements- formation of germ layers - metabolism during gastrulation - activity of gene during gastrulation - gastrulation in frog and chick. Cellular differentiation - cytodifferentiation and chemodifferentiation. Stem cells - totipotency and pluripotency. Embryonic stem cells and their applications.	15
IV	.Organogenesis- ectodermal derivatives - formation of central nervous system- development of eye in frog. Mesodermal derivatives - kidney in frog - endodermal derivative - intestine in frog. Regeneration - types of regeneration. Regenerative capacity in the animals – factors influencing regeneration – stimulation and suppression – polarity and gradients. Metamorphosis – types- amphibian metamorphosis- events and hormonal control. Insect metamorphosis -moulting - growth and hormonal control.	10

V	Hormones - hormones role in sexual cycles in mammals- placenta in mammals. Asexual reproduction - Assisted Reproductive Technology (ART) – male infertility – sperm abnormalities. Superovulation – *In vitro* fertilization (IVF) - Intracytoplasmic sperm injection (ICSI) -Gameto intrafallopian transfer (GIFT) – screening for genetic disorders. Surrogacy and ethical issues.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts and principles of developmental biology.	K1
CO2	Describe gametogenesis, fertilization and early embryonic development.	K2
CO3	Explain the processes of cell differentiation, morphogenesis and organogenesis	K3
CO4	Analyze the molecular and genetic mechanisms involved in growth and development.	K3
CO5	Apply the concepts of developmental biology to understand normal and abnormal development in organisms.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Barresi. M.J.F and Gilbert. S.F. (2020). Developmental Biology. OUP Publishers, USA.
- Arumugam.N. (2019). Developmental Biology. (II Edition). Saras Publications.
- Berry.A.K. (2016). An Introduction to Embryology. Emkay Publications, New Delhi.
- Reeves. L. (2022). Essential Developmental Biology. Murphy & Moore Publ
- Banerjee. S. (2001). Development Biology, Dominant Publishers & Distributors Pvt Ltd, New Delhi
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- Verma, P.S. And Agarwal V.K. (2005). Chordate Embryology (Developmental Biology). S. Chand& Company Ltd., New Delhi.
- Rastogi. V.B. (2012). Developmental biology. Kedar Nath Ram Nath Publ
- Schatten, H. and Schatten, G. (2012). Molecular Biology of Fertilization. Academic Press.
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- Browder, L.N. (1980). Developmental Biology. Saunders College, Philadelphia.
- Deuchar, E.M. (1976). Cellular Interaction in Animal Development. Chapman and Hall Publisher, London.

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SEMESTER II

First Year	CORE COURSE – IV (CC)	Credit : 5
Semester - II	IMMUNOLOGY	Sub Code: P25BTCC21

(Theory)

LEARNING OBJECTIVES

- To learn the types of Immunity
- To learn about the cells and organs involved in immune system
- To understand the diagnostic process of diseases by immunotechniques
- To learn about infection and immunity mechanism
- To understand the basic about vaccines

UNIT	CONTENTS	HOURS
I	The Immune System - Introduction - Cells of the Immune system – Innate a Acquired immunity - Primary and secondary lymphoid organs - Nature of antigens Chemical and molecular basis of antigenecity - Immunogenecity - Haptens - Adjuvar - Primary and Secondary Immune Responses - Theory of Clonal selection	10
II	Humoral Immunity - B-lymphocytes and their activation - Structure and function of Immunoglobulin - Isotypes of immunoglobulins - Antigen- Antibody interactions – Antibody affinity- avidity; Agglutination - Precipitation - Idiotypic antibodies - monoclonal antibodies - antibody engineering - Generation of antibody diversity - Major Histocompatibility Complex.	15
III	Cell Mediated Immunity - Biology of T lymphocyte - Classification of T lymphocytes - Structure of T Cell Receptor (TCR) - TCR diversity and genetics - Antigen presenting cells (APC) - macrophages - dendritic cells - Origin and functions of APC - Antigen processing and presentation - Cytokines - Cell mediated cytotoxicity - mechanism of T cell and NK cell mediated lysis - Complement- Hypersensitivity	15
IV	Immunity and Infection Mechanism- Tissue injury and InflammationImmunosuppression- Immunological Tolerance - Immunity to infectious agents - Transplantation- Autoimmunity - Tumor Immunology - Vaccines: Conventional- Molecular vaccines -Types of vaccines - Recent trends in Immunology of Infectious diseases.	10
V	Experimental Immunology – ABO Blood grouping and Rh typing, Immunodiffusion and Immunoelectrophoresis, Hemagglutination - production of polyclonal and monoclonal antibodies - Western Blotting – ELISA - Radio Immunoassay - FACS.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and components of the immune system.	K1
CO2	Describe the structure and functions of immune cells, organs and antibodies.	K2
CO3	Explain the mechanisms of innate and adaptive immunity and antigen-antibody interactions.	K3
CO4	Analyze immune responses involved in hypersensitivity, autoimmunity and transplantation	K3
CO5	Apply immunological principles and techniques in disease diagnosis, prevention and treatment	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

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- Elgert KD- 1996 Immunology: Understanding Immune system- John Wiley and sons.
- Roitt I- Brostoff- Male- 2001 Immunology- Mosby Publications.

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First Year	CORE COURSE – V(CC)	Credit : 5
Semester - II	GENETIC ENGINEERING	Sub Code: P25BTCC22

(Theory)

LEARNING OBJECTIVES

- Configuration of various vectors and cloning the gene of interest into them.
- Cloning a desired gene into a vector to produce a recombinant protein product in a heterologous expression system.
- Principles of various cloning strategies, transformation, screening, sequencing etc.
- Various tools of genetic engineering for developing mutants, diagnostics and other commercially important products
- Applications of Recombinant DNA technology in agriculture and healthcare.

UNIT	CONTENTS	HOURS
I	Importance and Outline of Recombinant DNA Technology, Organization of Genes in the Genome, Gene expression. Restriction Endonuclease- Type I, Type II, Type III, Type IV; S1 endonuclease, DNA ligase; Alkaline phosphatase; Reverse transcriptase; Polynulcleotide Kinase; Terminal transferase enzyme; Nuclease exonuclease, endonuclease, RNase and DNase; DNA polymerase; RNA Polymerase. Modifying enzymes, linker, Adaptor, Homopolymer tailing, DNA ligase.	10
II	Role of Promotors and Terminators, Cloning vectors for higher plants- Agrobacterium; Plant virus based vectors; cloning vectors for animal, viruses as cloning vectors for mammals. Vectors: Plasmid- Definition, Ideal characteristics of plasmids. Structure and uses of PBR322, PUC Vector- Lambda phage vector- Insertional and replacement vector, M13 Phage vector. Cosmid. Gene Transfer method- Transformation, Trnsduction, Particle bombardment, Electroporation, Liposome mediated gene transfer, Microjection, Agrobacterium mediated gene transfer.	15
III	Cloning cDNA in Plasmid vectors, cloning c DNA in bacteriophage vectors. Advanced cloning strategies, Multigene cloning and gateway clonings, Ligation independent cloning (LIC), Gibson DNA assembly. Cloning strategies- Construction- Genomic, r DNA Libraries, Probe construction, Methods of labeling gene probes- Recombinant selection and screening, Molecular cloning, Strategies for Identifying recombinant clones- Gene mapping technique. Selection Markers, Reporter genes used in Genetic engineering. Genome editing- TALENS, Zinc finger nucleases, CRISPR/CAS9.	15

IV	Cloning Strategies- Steps and strategies of gene cloning; Generation of Foreign DNA Molecules, Insertion of foreign DNA into vector- Recombinant DNA introduced into host cells- Microjection, Particle bombardment, Lipofection, Electroporation and Electrofusion. Analysis of cloned genes- Restriction enzymes analyses, Southern blotting, Northern blotting, Westerm blotting, colony & plague hybridization. **Selection Methods:** Genetic selection, Identification of recombinants- insertional inactivation, screening for recombinants lac selection, screening for blue white colonies, markers gene	**10**
V	Application of rDNA Technology- production of recombinant proteins- Insulin, Human growth hormone (HGH). Transgenic plants- insect resistance, disease resistance, transgenic animals- Molecular pharming.	**10**

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts, tools and techniques used in genetic engineering.	K1
CO2	Describe the structure and functions of vectors, restriction enzymes and cloning strategies.	K2
CO3	Explain recombinant DNA technology and gene transfer methods.	K3
CO4	Analyze the applications of genetic engineering in medicine, agriculture and industry.	K3
CO5	Apply genetic engineering techniques for gene cloning, expression and molecular analysis	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- An introduction to genetic engineering by Desmond ST Nicholl, Cambridge University Press, 1994.
- Genetics and Genetic Engineering 1st Edition is written by M. Follower.
- Genetic Engineering and Biotechnology: Concepts Methods and Applications. Edited by V.L. Chopra and Anwar Nasim. Published by Oxford and IBH, 1990.
- Methodology of Plant Genetics Engineering by. Rau, S. P. Likitha, 2018.
- Genetic Engineering and Its Applications" is written by Preeti Joshi, 2002.
- Introduction to biotechnology and genetic engineering. by: Nair, A. J. 2008.
- R.P. Meyyan, Genetics and Genetic Engineering, Saras Publication, 2002.
- Genetic engineering and biotechnology. V. L. Chopra, Anwar Nasim ; Oxford and IBH Pub. Co., New Delhi, 1990.

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First Year	CORE COURSE – VI (CC)	Credit : 5
Semester - II	PLANT AND ANIMAL BIOTECHNOLOGY	Sub Code: P25BTCC23

(Theory)

LEARNING OBJECTIVES

- To impart theoretical knowledge on various techniques of plant biotechnology like tissue culture, plant genetic transformation and their application in industries.
- Importance of secondary metabolites and production in plants.
- Use of molecular biology techniques genetically engineer the animals to improve sustainability, productivity and suitability for pharmaceutical and industrial applications.

UNIT	CONTENTS	HOURS
I	Definition Structure and organization of plant genome, Chloroplast and Mitochondrial genome, Totipotency. Plasticity and cytodifferentiation, Culture environment, Plant Tissue culture media and is components. Methods of sterilization, Different types of cultures - Shoot, callus cell suspension and protoplast cultures, Factors affecting in vitro culture, Organogenesis Micropropagation (Clonal Propagation), Somatic embryogenesis and synthetic	10
II	Embryo rescue techniques and importance of embryo culture. In vitro pollination and fertilization, Production of haploid plant through Androgenesis and Gynogenesis, Protoplast isolation and culture techniques, somatic hybridization and production of somatic hybrids and cybrids, Selection systems for somatic hybrids / cybrids and their characterization Cause and use of somaclonal variation, in vitro production of secondary metabolites cell suspension, root and hairy root cultures, Importance of germplasm conservation and methods.	15
III	Animal health disease diagnosis, hybridoma technique, monoclonal antibodies, application of probes for disease diagnosis of existing and emerging animal diseases. Prophylaxis - Vaccines, Oral vaccines DNA Vaccines in animal disease. Cell culture: primary and established culture; organ culture; tissue culture.	15
IV	Disaggregation of tissue and primary culture; cell separation, Slide and coverslip cultures, flask culture, test tube culture techniques, cell synchronization, cryo preservation. Scaling up of animal cell culture, cell line and cloning micromanipulation and cloning, somatic cell cloning. Karyotyping; measuring parameters for growth, measurement of cell death, apoptosis and its determination, cytotoxicity assays.	10
V	Stem cell culture and its types, Embryonic stem cells and their application Cryopreservation of gametes and embryos (Animal and Human), invitro fertilization and embryo transfer. Micro Insemination-PZD.ICSI, SUZI, MESA Transgenic animal: methods of gene delivery-Applications of transgenic animals, Safety regulation for transgenic animals Current issues related to transgenic animals	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic principles of plant and animal physiology and their importance in biological systems.	K1
CO2	Describe physiological processes in plants such as photosynthesis, respiration, transpiration, mineral nutrition, and plant growth regulation	K2
CO3	Explain physiological functions of animal organ systems including digestion, respiration, circulation, excretion, and nervous coordination	K3
CO4	Analyze the mechanisms regulating growth, development, reproduction, and homeostasis in plants and animals.	K3
CO5	Evaluate physiological responses of plants and animals to environmental factors and their adaptive significance.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Razdan.M.K.,2011.Planttissueculture.OxfordandIBHpublishingCompanyPvt. Ltd, New Delhi.
- Chawla. H. S., 2010. Introduction to plant biotechnology. Oxford and IBH publishingcompany pvt. Ltd, New delhi.
- IanFreshney,2010.Cultureofanimalcells.6thedition,Wiley-Blackwellpublishers.

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First Year	CORE PRACTICAL – II (CP)	Credit : 4
Semester - II	IMMUNOLOGY, PLANT AND ANIMAL BIOTECHNOLOGY, GENETIC ENGINEERING (P)	Sub Code:P25BTCP2

(Practical)

LEARNING OBJECTIVES

- To learn about the basic principles and techniques involved in Immuno technology
- Develop skills to determine antigen and antibody in the sample by various methods.
- Perform blood typing assays and differentiate between blood cells.

PRACTICALS

IMMUNOLOGY

1. Blood Group – typing
2. Blood Cell Counting - RBC and WBC counting
3. Immuno electrophoresis
4. Immuno diffusion
5. Rocket Immuno electrophoresis
6. ELISA
7. Western Blotting (Demo)
8. Separation of proteins on SDS-PAGE

GENETIC ENGINEERING

1. Preparation of plasmid DNA by alkaline lysis method.
2. Cloning of fragments in PBR322
3. RAPD
4. RFLP
5. Amplification of DNA - PCR

6. Determination of molecular weight of DNA

PLANT AND ANIMAL BIOTECHNOLOGY

1. Plant tissue culture media preparation
2. Isolation of plant protoplast
3. Culture of plant protoplast
4. Localization of nucleus using nuclear stain
5. Agrobacterium culture maintenance and isolation of plasmid DNA.
6. Mass culture of Chlorella /Spirulina
7. Introduction to Animal Cell culture: Procedure for handling cells and medium.
8. Preparation of tissue culture media
9. Preparation of single cell suspension from chicken liver (Primary cell culture).
10. Trypsinization of established cell culture.
11. MTT Assay

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Identify and use laboratory instruments, reagents and safety procedures related to immunology, genetic engineering and biotechnology experiments.	K1
CO2	Describe the principles and methodologies involved in immunological, genetic engineering and biotechnology techniques	K2
CO3	Perform basic immunological assays, recombinant DNA techniques, plant tissue culture and animal cell culture methods	K2
CO4	Analyze experimental data obtained from immunological, molecular and biotechnological studies.	K4
CO5	Apply practical skills in immunology, genetic engineering and plant and animal biotechnology for research and industrial applications.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References

- J.H. Miller (1992), A short course in Bacterial Genetics by Cold Spring Harbor Laboratory.
- DM. Glover and BD. Hames (1995), DNA cloning I and II by IRL Pres.
- MA. Innis- DH- GelfandandD JJ Sninskey (1995), PCR strategies by Acadmic Press.
- Molecular cloning Volume I- Volume II and Volume III- Academic Press.

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First Year	ELECTIVE COURSE – II (EC)	Credit : 4
Semester - II	A.MEDICAL BIOTECHNOLOGY	Sub Code: P25BTEC2A

(Theory)

LEARNING OBJECTIVES

- Students will learn about the genetic, cellular, and molecular mechanisms underlying various diseases.
- This includes areas like genetic engineering, biopharmaceuticals, diagnostic techniques, and regenerative medicine.
- The course will cover the development and application of diagnostic tools for disease detection and the development of therapeutic strategies, including gene therapy and stem cell therapies.
- The program will emphasize the importance of lifelong learning and staying updated with the latest advancements in the field.

UNIT	CONTENTS	HOURS
I	Gross and microscopic anatomy, Functional and comparative anatomy, Neuroanatomy and developmental biology, Physiology of major systems (cardiovascular, respiratory, renal, endocrine, nervous, etc.),	10
II	Structure and function of cells and organelles, Cell signaling, cell cycle, and apoptosis, DNA/RNA structure, replication, transcription, translation, Gene regulation and expression, Biomolecules: proteins, carbohydrates, lipids, nucleic acids, Enzyme kinetics and regulation, Metabolic pathways and disorders.	15
III	Mendelian and molecular genetics, Human genome project, genetic disorders, Genomic and proteomic analysis, Pathogenic microorganisms and mechanisms of disease, Host-pathogen interactions, Immune system structure and function, Immunoassays, monoclonal antibody production, cytokines	15
IV	General pharmacological principles, Systemic pharmacology (drugs affecting major organ systems), Toxicology, and antidotes	10
V	Spectrophotometry, chromatography, electrophoresis, Cell and tissue culture, microscopy, Protein purification, immunoassays, molecular diagnostics, Bioinformatics and computational biology.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts and applications of medical biotechnology in healthcare	K1
CO2	Describe the principles of molecular diagnosis, therapeutic products and biomedical technologies	K2
CO3	Explain the role of genetic engineering, stem cell technology and recombinant products in medicine	K3
CO4	Analyze the applications of biotechnology in disease diagnosis, prevention and treatment.	K3
CO5	Apply medical biotechnology techniques for healthcare, pharmaceutical and clinical research applications	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Medical Microbiology" by Murray et al.
- Clinical Chemistry: Principles, Techniques, and Correlations" by Bishop et al.
- Drug Discovery: A History" by Walter Sneader
- Principles of Pharmacology" by Golan et al.
- Fundamentals of Medical Biotechnology" by Rajneesh Prajapat
- Medical Biotechnology" by Oxford University Press
- Genetics: Analysis of Genes and Genomes" by Griffiths et al.
- Human Molecular Genetics" by Strachan and Read
- Lewin's Essential Genes" by J.D. and Jocelyn E. Krebs
- Textbook of Cell and Molecular Biology. Books and Allied Ltd, Ajoy Paul. 2014.
- Nucleic Acids in Chemistry and Biology. Oxford University Press, Blackburn GM and Gait MJ. 2006.
- Fundamentals of Medical Biotechnology" by Rajneesh Prajapat,
- Ananthanarayan and Paniker's Textbook of Microbiology",
- Journals: Nature Biotechnology, Trends in Biotechnology, Biotechnology Advances, Journal of Biotechnology.

- Online Databases: NCBI Bookshelf, R2 Digital Library
- Online Courses: Coursera, edX, FutureLearn
- https://library.hccs.edu/med-biotech/oer
- https://www.nlm.nih.gov/
- https://www.ncbi.nlm.nih.gov/guide/all/

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First Year	ELECTIVE COURSE – II (EC)	Credit : 4
Semester - II	B.NANO BIOTECHNOLOGY	Sub Code: P25BTEC2B

(Theory)

LEARNING OBJECTIVES

- Gaining a deep understanding of the unique physical and chemical properties of nanomaterials and how these differ from their bulk counterparts.
- Learning how nanomaterials interact with biological molecules and systems at the cellular and molecular level.
- Investigating the application of nanobiotechnology in cancer diagnosis, treatment, and theranostics Applying knowledge and skills to solve complex problems in nanobiotechnology research and development

UNIT	CONTENTS	HOURS
I	Introduction to Nanotechnology- Scientificrevolution, Feynman's vision, Classification of nanobiomaterials -Types of nanomaterials – nanoparticles, nanotubes, nanowires, Nanofibers, Nature'sNano phenomena.	10
II	Preparation of Nanomaterial - Top down and bottom up approaches, Biosynthesis, Nanobiomaterials- Polymer, Metal based Nanobiomaterials, Carbon based Nanomaterials, DNA based Nanostructures, Protein based Nanostructures, Quantum dots, Magnetic Nanoparticles, Nanofibres, Hydrogels, Films and Scaffolds.	15
III	Green synthesis and characterizationanalysis of nano materials, spectroscopy - optical techniques like UV-vis and fluorescence spectroscopy, FTIR spectroscopy. Microscopy- Electronmicroscopy -TEM ,SEM,AFM microscopsy, dynamics light scattering,zeta potential measurement,XRD, SPR and SERS.	15
IV	Nanomaterials for diagnosis and therapy, Implications and NPs based drug delivery,Nano -carriers for application in medicine, polymeric nanoparticles as drug carriers, Drug release mechanism, Targeted Drug Delivery using nanocarriers, Nanoparticle technologies for cancer therapy and diagnosis, Point of Care and Personalized medicine, Magnetic nanoparticles for imaging and Hyperthermia.Risk assessment and safety regulations of nanoparticles.	10
V	Nanotoxicology, Portals of Entry of the nanoparticles into the Human Body, Bio-toxicity of Nanoparticles, Nanoparticles in Mammalian systems and Health threats, Biological response and cellular interaction of implant materials and scaffolds, application of nano materials in bone substitutes and dentistry, food and cosmetic application,Bio-sensors and lab on a chip,Bio-devices and implantable devices,Nano materials for anti-microbial coating-medical implants and paints, application of nanotechnology in textile industry	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and principles of nanobiotechnology.	K1
CO2	Describe the synthesis, characterization and properties of nanomaterials and nanoparticles.	K2
CO3	Explain the applications of nanobiotechnology in medicine, agriculture and environmental science.	K3
CO4	Analyze the role of nanomaterials in drug delivery, diagnostics and biosensor development	K3
CO5	Apply nanobiotechnological techniques for the development and evaluation of nanomaterial-based products.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Nanotechnology, S.Shanmugam, Mjppublication. 2011.
- Advanced nanomaterials, kurt E.geckeler, HiroyukiNishide ,WileyVHC.2010.
- Nanotechnology and tissue engineering. T.Laurencin,LakshmiS. Nair,CRCpress.2012.
- Hand book ofcarbon nanomaterials. FrancisD souza,KarlM. Kadish.
- World scientific publishing co.pte. ltd. 2011.
- Oded Shoseyov (Editor),IlanLevy,2010.NanoBioTechnology:BioInspiredDevicesand Materials of the Future, Humana Press.
- Chad A. Mirkin and ChristofM. Niemeyer,2007. NanobiotechnologyII: MoreConcepts and Applications, Wiley-VCH.
- Challa S.S.R.Kumar (Ed). 2006. Biologicals and pharmaceutical nanomaterials, Wiley-VCH Verlag Gmbh& Co, KgaA.
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First Year	NON MAJOR ELECTIVE – I (NME)	Credit : 2
Semester - II	MOLECULAR BIOTECHNOLOGY	Sub Code: P25BTNM1A

(Theory)

LEARNING OBJECTIVES

- Students will learn how to apply molecular biology principles to solve problems in diverse fields like medicine, agriculture, and industrial biotechnology.
- Core competencies in critical thinking, experimental design, and data analysis, as well as fostering the ability to communicate scientific research effectively.

UNIT	CONTENTS	HOURS
I	History, scope, and importance of biotechnology, Structure and function of DNA and RNA, Nucleic acids as genetic material.	10
II	DNA sequencing, DNA cloning and plasmid construction, Electrophoresis (DNA, protein), CRISPR genome editing, SDS-PAGE and Western blotting, Cell and Tissue Culture, Plant and animal cell culture techniques	15
III	DNA modifying enzymes and vectors, Methods of recombinant DNA technology, Nucleic acid hybridization, Gene libraries, PCR amplification	15
IV	Antigen-antibody interactions, Immunoassays (ELISA, immunodiffusion), Sequence alignment and phylogenetic analysis, Database use, Protein structure analysis.	10
V	Genomics and Proteomics, Genomic sequencing and analysis, Proteomic profiling and 2D gel electrophoresis, Bioinformatics tools for sequence analysis. Biostatistics and Research Methodology- Statistical tests, Research design and data analysis.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and principles of molecular biology.	K1
CO2	Describe the structure, organization and functions of DNA, RNA and proteins.	K2
CO3	Explain the processes of DNA replication, transcription and translation	K2
CO4	Analyze gene regulation and molecular mechanisms involved in cellular functions.	K3
CO5	Apply molecular biology techniques for genetic analysis, research and biotechnology applications	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

1. Molecular Biology of the Gene" by James D. Watson, Tania A. Baker, Stephen P. Bell, Alexander Gann, Michael Levine, and Richard Losick:
2. Molecular Cell Biology" by Harvey Lodish, Arnold Berk, S. Lawrence Zipursky, Paul Matsudaira, David Baltimore, and James Darnell:
3. Molecular Biology of the Cell" by Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, and Peter Walter
4. Lewin's Essential Genes" by J.D. and Jocelyn E. Krebs
5. Textbook of Cell and Molecular Biology. Books and Allied Ltd, Ajoy Paul. 2014.
6. Nucleic Acids in Chemistry and Biology. Oxford University Press, Blackburn GM and Gait MJ. 2006.
7. Cell and Molecular Biology. 8th Edition. Saunders Company, De Robertis, EDP and E.M.F Robertis. 1987.
8. Pollard, T. D., Earnshaw, W. C., Lippincott-Schwartz, J., & Johnson, G. (2016).
9. Essential Cell Biology" by Bruce Alberts, Dennis Bray, Karen Hopkin, Alexander D. Johnson, Julian Lewis, Martin Raff, Keith Roberts, and Peter Walter
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First Year	NON MAJOR ELECTIVE - I (NME)	Credit : 2
Semester - II	HEALTHCARE TECHNOLOGY	Sub Code: P25BTNM1B

(Theory)

LEARNING OBJECTIVES

- To develop and implement a capacity building initiative in healthcare technology for healthcare professionals
- To impart basic knowledge of healthcare technology and its practical application
- To provide awareness of technology application in patient and functional management
- To prepare for technology led advancements and innovations in healthcare along with regulatory and practical aspects

UNIT	CONTENTS	HOURS
I	Introduction to human body, Definition and subdivision of anatomy of Anatomical position and terminology. Cell Structure, organelles, inclusion bodies, nucleus, Structure of chromosome, DNA, RNA. Basics & fundamentals of Genetics, Karyotyping, Chromosomal disorders, prenatal diagnosis, genetic counseling	10
II	Cardio vascular system - Types and general features of blood vessels. Structure and types of arteries and veins. Blood supply of the heart. Lymphatic system - General features of Lymph node and lymphatic vessels. Respiratory system - Name the organs of respiration. Mention the microscopic feature of lung. Digestive system - Name the parts of the alimentary canal and accessory organs. Esophagus, stomach, small and large intestine. Salivary glands, pancreas, liver and gall bladder and liver. Urinary system - Names of urinary organs. Location and features of kidney, ureter, urinary bladder & urethra. Microscopic feature of kidney. Reproductive system - Male and female organs of reproduction.	15
III	Nervous system, Spinal cord-location, extent, external features and blood supply Cranial nerves-name, number, attachment, area of distribution, Spinal nerves-typical spinal nerve. Location and distribution of brachial and lumbosacral plexus. Autonomic nervous system-sympathetic and parasympathetic nervous system. Location of pre-ganglionic and post-ganglionic neurons. Sense organs, Location and features of nose, tongue, eye, ear and skin. Endocrine system.	15
IV	Clinical decision support systems. (CDSS), Artificial intelligence and algorithms in Healthcare, Patient monitoring systems (PMS), Role of IoT and Point of Care devices in Healthcare. Technology-led Health Care, Quality Management using Technology. Technology for infection prevention	10

	and waste management: Development of medical devices and ensuring quality of medical devices. Tissue Engineering Case Study- 3D printing in healthcare, augmented reality and virtual reality in healthcare, Robotics and biomechanics in healthcare.	
V	Healthcare technology - Regulatory, policy and Practical aspects .Legal, ethical and Intellectual property rights in Healthcare technology and Cyber security in hospitals, Health technology policies and guidelines. Technology and its practical application, Understanding health care data and standards, National Digital Health Mission.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts and significance of healthcare biotechnology.	K1
CO2	Describe the applications of biotechnology in disease diagnosis, prevention and treatment.	K2
CO3	Explain the role of vaccines, recombinant therapeutics and molecular diagnostics in healthcare.	K2
CO4	Analyze the impact of biotechnological advancements on public health and personalized medicine.	K3
CO5	Apply healthcare biotechnology principles for clinical, pharmaceutical and research applications.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Princy Louis Palatty et.al - A Textbook of Bioethics for Healthcare Professionals, 1st Edi, 2018
- Ross and Wilson Anatomy and Physiology in Health and illness, Elsever, 13th Edi, 2018.
- Chaurasia, Human Anatomy, CBS Publisher, 5th Edi, Vol 1 to 3, 2010.
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- The Innovator's Prescription: A Disruptive Solution for Health Care 1st Edition by Clayton M. Christensen , Jerome H. Grossman M.D., Jason Hwang M.D.
- Anatomy, Physiology Health Education: Rohini Agrawal,Neeraj Agrawal.CBS Publishers
- Principles of Human Physiology, Surrinder H Singh, CBS Pubisher
- Undersstanding Psychology, Robert S.Feldman
- Healthcare Technology Management –A systematic Approach, Francis Hegarty, John Amoore, Paul Blackett, Justin McCarthy, Richard Scott, CRC Press Publisher.

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SEMESTER III

Second Year	CORE COURSE – VII (CC)	Credit : 5
Semester - III	RESEARCH METHODOLOGY	Sub Code: P25BTCC31

(Theory)

LEARNING OBJECTIVES

- To identify and analyze the research problem systematically
- To collect and interpret accurate data using suitable research methods

UNIT	CONTENTS	HOURS
I	An Introduction: Meaning ofResearch, Objectives of Research, Types of Research;Research Approaches, Importance of knowing howresearch is done, Research Process, Criteria of good research. Defining the Research Problem; Research Design; Sampling Design; Methods of DataCollection; Processing and Analysis of Data;Sampling Fundamentals.	10
II	Review of literature, Writing the Research Report(Thesis and publications): Components of researchreport - Title, Authors, Addresses, Abstract,Keywords, Introduction, Materials and Methods,Results, Discussion, Summary, Acknowledgements and Bibliography.	15
III	Standard Deviation- T test. Analysis of Variancecomponents (ANOVA) for fixed effect model; Total,treatment and error of squares, Degrees of freedom,Confidence interval; ANOVA for random effectsmodel, Estimation of variance components, Modeladequacy checking. Two factor Factorial Design.	15
IV	Spreadsheet Tool: Introduction to spreadsheetapplication, features and functions, Using formulasand functions, Data storing, Features for Statisticaldata analysis, Generating charts/ graph and otherfeatures. Presentation Tool: Introduction topresentation tool, features and functions, Creatingpresentation, Customizing presentation, Showingpresentation. Tools used may be Microsoft PowerPoint, Open Office or similar tool	10
V	Web Search: Introduction to Internet, Use of Internet and WWW, Using search engine like Google, Yahoo, Pubmed, Science direct, Scopus etc, and Using advanced search techniques	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts, principles, and importance of research methodology in scientific studies.	K1
CO2	Describe various research designs, methods of data collection, and techniques used in research.	K2
CO3	Apply appropriate research methods and statistical tools for conducting scientific investigations.	K3

CO4	Analyze research data and interpret findings for drawing meaningful conclusions.	K3
CO5	Demonstrate skills in research proposal writing, scientific reporting, and ethical practices in research.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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- Authoring a PhD, thesis: how to plan, draft, write and finish a doctoral
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Second Year	CORE COURSE – VIII (CC)	Credit : 5
Semester - III	BIOPROCESS TECHNOLOGY	Sub Code: P25BTCC32

(Theory)

LEARNING OBJECTIVES

- Give an idea about the avenues of improving the trait of microbes and study the downstream processes for product recovery in fermentation
- Create adequate industrial knowledge and skills

UNIT	CONTENTS	HOURS
I	Introduction to fermentation. General requirements of fermentation. Microbial growth kinetics of batch andcontinuous culture. Solid substrate, slurry fermentation and its application. Microbial cell culture. Immobilization of cells and enzymes. Food Safety: Introduction to food safety aspects and food related hazards – HACCP and ISO.	10
II	Types of bioreactors: Submerged reactors, surface reactors, mechanically agitated reactors, non- mechanically agitated reactors. Design of fermenters, body construction. Production of citric acid, penicillin and insulin. Isolation and improvement of Industrially important Micro-organisms, Media for Industrial fermentation and Sterilization.	15
III	Introduction to bioproducts and bioseparation. Primary recovery process: Cell disruption methods. Cell lysis and Flocculation: Osmotic and mechanical methods of lysis. Flocculation, Filtration: Principles, Conventional, Crossflow filtration. Sedimentation: Principles, Sedimentation coefficients. Extraction Principles, Liquid liquid extraction, aqueous two phase extraction, supercritical fluid extraction.	15
IV	Down Stream Processing: Chromatography Techniques, Membrane separation, ultrafiltration. Drying: Principles and operation of vacuum dryer, shelf dryer, rotary dryer, freezer and spray dryer. Crystallization and Whole broth processing	10
V	Aerobic and anaerobic fermentation processes and their application in the field of biotechnology industry. Production of commercially important primary and secondary metabolites, Effluent Treatment and Fermentation Economics.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and principles of Bioprocess Technology and its industrial applications.	K1
CO2	Describe the components and operation of bioprocess systems including bioreactors, fermentation, and downstream processing.	K2
CO3	Apply bioprocess techniques for the production of biological products and industrial metabolites.	K3
CO4	Analyze bioprocess parameters and evaluate factors affecting product yield and process efficiency.	K3
CO5	Demonstrate knowledge and practical understanding of bioprocess operations, process optimization, and quality control practices.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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Second Year	CORE COURSE – IX (CC)	Credit : 5
Semester - III	ENVIRONMENTAL BIOTECHNOLOGY	Sub Code: P25BTCC33

(Theory)

LEARNING OBJECTIVES

- The students will understand the impact of pollution on the environment and the need for remediation with the use of microbes and biological methods and they also learn about the biorenewal fuels.

UNIT	CONTENTS	HOURS
I	Environment: Basic concepts and issues; Environmental management and Conservation, Environmental Laws & Agencies involved in conservation. Environmental Pollution: Types of pollution & its control strategies -Air pollution, Soil pollution, Water pollution, Oil pollution & Radioactive pollution	10
II	Biofilm Kinetics: Completely mixed biofilm reactor- Soluble microbial products and inert biomass-Special- case biofilm solution. Reactor types:- batch reactor - continuous-flow stirred-tank reactor- Plug-flow reactor. Engineering design of Reactors in series.	15
III	Waste water management, source of waste water, Waste water treatment-physical, chemical and biological treatment. Microbiology of Waste water; Aerobic and anaerobic process, BOD and COD.	15
IV	Toxicity: Types and Test for evaluating Toxicity. Biosensors, Biomonitoring of toxic materials. Biomagnification, Biomining and Biofuels.	10
V	Bioremediation; In-situ and Ex-situ Bioremediation of contaminated soils and waste land; Microbiology of degradation of Xenobiotics in environment; Pesticides, Surfactants, Degradative plasmids. Solid waste: Composting, Vermiculture and methane production	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and principles of Environmental Biotechnology and its applications in environmental management.	K1
CO2	Describe the role of microorganisms and biotechnological processes in pollution control, waste treatment, and environmental sustainability.	K2

CO3	Apply biotechnological techniques for bioremediation, wastewater treatment, and environmental monitoring.	K2
CO4	Analyze environmental problems and evaluate biotechnological approaches for pollution prevention and ecosystem management.	K4
CO5	Demonstrate knowledge and practical understanding of environmental conservation, sustainable technologies, and eco-friendly biotechnological applications.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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Second Year	CORE COURSE – III (CP)	Credit : 4
Semester - III	BIOPROCESS TECHNOLOGY, ENVIRONMENTAL BIOTECHNOLOGY(P)	Sub Code: P25BTCP3

(Theory)

LEARNING OBJECTIVES

- To understand the basic techniques usen in bioprocess technology
- To understand biotechnological techniques for environmental protection and sustainability

Practicals:

BIOPROCESS TECHNOLOGY

1. Parts and Design of Fermentor
2. Solid State Fermentation
3. Submerged Fermentation
4. Foaming and Antifoaming agents
5. Media Preparation and Sterilization
6. Isolation of Industrially important microorganisms for microbial processes
7. Conservation of Bacteria by Lyophilization
8. Production and Estimation of Protease
9. Production and Estimation of Amylase
10. Production of Wine using Grapes

ENVIRONMENTAL BIOTECHNOLOGY

1. Isolation and enumeration of microorganisms from soil samples
2. Isolation and identification of Water borne microorganisms
3. Determination of Ph anf Dissolved oxygen (DO) in water samples
4. Estimation of Biological oxygen demand (BOD) and Chemical oxygen demand in waste water.

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles and applications of bioprocess technology and environmental biotechnology.	K1
CO2	Explain the preparation, sterilization and operation of bioprocess equipment used in laboratory practices.	K2
CO3	Perform microbial cultivation, fermentation and downstream processing techniques under laboratory conditions.	K3

CO4	Analyze environmental samples and evaluate biotechnological methods used for pollution control and waste management.	K3
CO5	Demonstrate practical skills in bioprocess operations, environmental monitoring and interpret experimental results effectively.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Encyclopedia of Industrial Biotechnology: Bioprocess, Bioseparation, and Cell Technology, Edited by Michael C. Flickinger Wiley, New York, Volumes 1-7, 2010,
- 2. M. S. Clark. 1997. Plant Molecular Biology: A Laboratory Manual. Springer-Verlag.
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Second Year	ELECTIVE COURSE – III (EC)	Credit : 4
Semester - III	A. FOOD BIOTECHNOLOGY	Sub Code: P25BTEC3A

(Theory)

LEARNING OBJECTIVES

- This is helps to the students to understand about food preparation
- To know about Food borne disease
- To help the students to understand about the industrial food process

UNIT	CONTENTS	HOURS
I	Introduction to Food Science and Technology: Foodchemistry, carbohydrates, proteins, lipids, vitamins,minerals. Nutrition, calorific value of foods, Diet and different types of diet.	10
II	Standards for food analysis; identity, purity and methods (cereals, vegetables, fruits) Food processing and preservation.	15
III	Fundamentals of Food Biotechnology; Role of biotechnology in fermented food products (dairy products, oriental fermentations, alcoholic beverages,and food ingredients), Enzymatic processing of fruit juices. Role of enzymes in baking and meat processing.	15
IV	Food Safety; Food Contaminants (Microbial,Chemical, Physical), Food Adulteration (Commonadulterants), Food Additives (functional role, safetyissues	10
V	Biotechniques; Applications of immunological techniques to food industry; Detection methods for pathogens from food samples, pesticide Residues, Newer sources of Ingredients.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles, scope and importance of Food Biotechnology in food production and human welfare	K1
CO2	Explain the role of microorganisms, enzymes and fermentation processes in food biotechnology.	K2
CO3	Apply biotechnological techniques for the production, preservation and quality improvement of food products.	K3

CO4	Analyze the applications of genetic engineering and molecular techniques in food processing and safety.	K3
CO5	Evaluate food quality, safety standards and recent advancements in food biotechnology for sustainable food development.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- B.Siva 2011.Food Processing & Preservation –PHI Learning Pvt Ltd.
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- D.G. Rao, 2010. Fundamentals of Food Engineering –PHI Learning Pvt Ltd.
- 2. Narang, Food Microbiology
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Second Year	ELECTIVE COURSE – III (EC)	Credit : 4
Semester - III	MICROBIAL BIOTECHNOLOGY	Sub Code: P25BTEC3B

(Theory)

LEARNING OBJECTIVES

- Teach the significance of microbes and their biotechnological applications in various sectors
- Impact the knowledge about the genetic improvement of microbes for human

UNIT	CONTENTS	HOURS
I	General concepts of microbial biotechnology. Microorganisms as factories for the production of novelcompounds. Genetic engineering of microbes to improve production of antibiotics, amino acids, lipids,enzymes, steroids and secondary metabolites. Biopolymers and bioplastics.	10
II	Definition, Concepts- history, biotechnological potentials of microalgae – food, feed Colourant, fuel and pharmaceutically valuable compounds. Cultivation methods of algae with reference to Dunaliella. Production of microbial biofertilizers cyanobacteria, Rhizobium, Azotobacter, Azospirillum, Phosphobacteria and VAM.	15
III	Biological pest control, scheme for selection of microbial antagonist for biological control of insects,bacterial, fungal and viral diseases. Mode of action of biological control involved in different biocontrolagents. Genetics of antimicrobial metabolite production in biocontrol bacteria. Risks associated with GMOs, Potential impacts on the environment and human health.	15
IV	Bioconversion of cellulosic and non-cellulosic wastes. Agrobyproducts, Biopulping, Biobleaching.Bioremediation of wood, lubricants, rubber and plastics. Biofuels.	10
V	Waste utilization: Waste water treatment - Aerobic and Anaerobic processes, Treatment schemes forwaste waters of dairy, distillery, tannery, sugar, antibiotic industries. Sewage disposal, compost making,methane generation. Microbiology of degradation of xenobiotics in environment: Ecologicalconsiderations, decay behavior, hydrocarbons, substituted hydrocarbons, oil pollution, surfactants, pesticides and dyes.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles of microbial biotechnology and explain the role of microorganisms in industrial, agricultural, environmental, and medical applications	K1
CO2	Describe the structure, physiology, genetics, and metabolic activities of industrially important microorganisms.	K2
CO3	Explain the techniques used for isolation, screening, cultivation, and preservation of microorganisms for biotechnological applications.	K3
CO4	Analyze microbial processes involved in the production of enzymes, antibiotics, biofuels, biofertilizers, and other value-added products	K3
CO5	Demonstrate microbial biotechnology techniques, including fermentation, bioprocessing, and downstream processing for product development.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Nelson, K. E. (2015). Encyclopedia of Metagenomics. Genes, Genomes and
- Metagenomes: Basics, Methods, Databases and Tools. Boston, MA: Springer
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- Bernad R. Glick and Jack J. Pasternak. Molecular Biotechnology Principles
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Second Year	NON MAJOR ELECTIVE – II (NME)	Credit : 2
Semester - III	A.AGRICULTURAL BIOTECHNOLOGY	Sub Code: P25BTNM2A

(Theory)

LEARNING OBJECTIVES

- Introduce the basics, civilization and development of agriculture
- Explain the concepts of agricultural crop improvement
- Understand the importance of agricultural biotechnology and laws

UNIT	CONTENTS	HOURS
I	Principles of Biotechnological Application in Agriculture , Introduction to microbialworld , Prokaryotic and eukaryotic microbes. Microbial growth in models of bacterial, yeast and mycelia growth curve.	10
II	Bacteria: cell structure, chemoautotrophy, photo autotrophy, growth. Bacterial genetics:Genetic recombination transformation, conjugation and transduction, plasmids, transposon	15
III	Role of microbes in soil fertility and crop production , Carbon, Nitrogen, Phosphorus and Sulphur cycles ,Biological nitrogen fixation- symbiotic, associative and asymbiotic. Azolla, bluegreen algae and mycorrhiza. Rhizosphere and phyllosphere.	15
IV	Microbes in human welfare and silage production, biofertilizers, biopesticides, biofuelproduction and biodegradation of agro-waste	10
V	Sterilization methods – Physical and chemical, Isolation of pure cultures and preservation of cultures. Mushrooms- edible and poisonous types, nutritive values, Culturing and production technique	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles, scope, and importance of Agricultural Biotechnology in improving crop productivity and sustainability.	K1
CO2	Describe plant tissue culture techniques, genetic engineering methods, and their applications in agriculture	K2
CO3	Apply biotechnological tools for crop improvement, disease resistance, and stress tolerance	K3
CO4	Analyze the role of molecular markers, transgenic crops, and biofertilizers in modern agriculture.	K3
CO5	Evaluate the benefits, limitations, biosafety, and ethical issues associated with agricultural biotechnology	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Jones L. 1991. Biotechnological innovation in Crop improvement,
- ButterworthHiemann, London
- Arie Altman. 1998. Agricultural Biotechnology, Marcel Dekker, Inc.
- Forbes JC and Watson RD. 1992. Plants in Agriculture, Cambridge Univ.
- Press, Great Britain
- Erbish FH and Maredia M. 1998. Intellectual Property Rights in Agricultural
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Second Year	NON MAJOR ELECTIVE – II (NME)	Credit : 2
Semester - III	B.MARINE BIOTECHNOLOGY	Sub Code: P25BTNM2B

(Theory)

LEARNING OBJECTIVES

- To identify and characterize the flora of marine ecosystem
- To identify, characterize and understand and life history of marine fauna
- To explore the assessment of marine bioresources and protect marine environment

UNIT	CONTENTS	HOURS
I	Biotechnology in Marine Sciences. Aquaculture: culture of shrimp, crab, edible mollusc, oysters and pearl oysters, Culture of milkfish, mullets and eel. Culture of sea weeds.Culture of live feed organisms- brine shrimp, rotifers.Marine micro algae- aquaculture, antioxidants-carotenoids, astaxanthin	10
II	Marine hydrocolloids-agar,agarose, carageenan, alginates, chitosans and chitin. Marine enzymes - Applications of enzyme for fish processing. Marine Lipids-application of lipases for modification of fats and oils. Marine flavourants. Bioconversion of organic materials and fish ensilage.	15
III	Aquaculture biotechnology- hormonal manipulation of sex,chromosomal manipulation of sex fish,cryopreservation of fish gametes and embryo. Diseases of cultured shrimp, fish. Diagnostics and theirapplication to aquaculture.	15
IV	Production of transgenic fishes-growth hormone, antifreeze protein, disease resistant fish, application of hormones in induced breeding in aquaculture. Antifreeze protein and its applications.	10
V	Pharmaceuticals from marine realms, type of drugs from marine organisms and their medical applications. Biofouling and their control. Marine bioremediation-Biosurfactants and Control of oil spills. Extremophiles	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles, scope, and importance of Agricultural Biotechnology in improving crop productivity and sustainability.	K1
CO2	Describe plant tissue culture techniques, genetic engineering methods, and their applications in agriculture	K2
CO3	Apply biotechnological tools for crop improvement, disease resistance, and stress tolerance	K3
CO4	Analyze the role of molecular markers, transgenic crops, and biofertilizers in modern agriculture.	K3
CO5	Evaluate the benefits, limitations, biosafety, and ethical issues associated with agricultural biotechnology.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Biology of Fishes 2nd Edn, 1996, Carl E. Bond, W.B. Saunders Company, Philadelphia.Mapping the Diversity of Nature, 1994, Miller RI, Chapman &Hall.
- Global Biodiversity Assessment. UNEP, 1995, Heywood V.H, Cambridge University Press.
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- Ecology and Biodiversity of Indian Mangroves, 1999, Naskar K and Mandal R, Daya Pubishing corporation.
- Marine Biology: Function, biodiversity, ecology, 2001, Jeffrey S., Levinton, Oxford University Press.
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SEMESTER IV

Second Year	CORE COURSE – X (CC)	Credit : 5
Semester - IV	BIOINFORMATICS AND BIOETHICS	Sub Code: P25BTCC41

(Theory)

LEARNING OBJECTIVES

- Understand basics of bioinformatics which includes recent advancement in computer application
- Analyze the biological data using bioinformatics tools and software
- Know about specific application of software and algorithms used for the clear understanding of biological data

UNIT	CONTENTS	HOURS
I	Introduction to bioinformatics, Database (NCBI), Proteinand nucleotide databases, Information retrieval from biological databases, Sequence alignment and database searching-similarity searches using BLAST and FASTA	10
II	Sequence alignment basics, match, mismatch, similarity, gap penalty, proteinvs DNA alignments, pairwise alignment. Global and local alignment algorithms ,multiple sequence alignment: Tools of Multiple sequence alignment	15
III	Bioinformatics for genome sequencing, Regulatory sequence analysis, Bioinformatics for Genome maps and markers, Bioinformatics for understanding Genome variation, Protein structure-X-ray crystallography, SCOP,CATH, DALI and HSSP; Visualization of molecular structures-RasMol and Pymol; Protein secondary structure prediction.	15
IV	Overview of ethical issues in biological sciences - interference with nature, bioethics in health care -patient confidentiality, informed consent, euthanasia, artificial reproductive technologies, prenatal diagnosis, genetic screening, gene therapy and transplantation.	10
V	Bioethics in healthcare, agriculture, modern biology, biotechnology.Bioethics in stem cell research, Human and animal experimentation, Animal rights/welfare.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamentals, scope, and applications of Bioinformatics and Bioethics in life sciences	K1
CO2	Describe biological databases, sequence analysis tools, and ethical principles related to biotechnology research	K2
CO3	Apply bioinformatics software and online resources for retrieval and analysis of biological data.	K3
CO4	Analyze genomic, proteomic, and phylogenetic data using bioinformatics approaches and evaluate ethical issues in biotechnology	K3
CO5	Assess the importance of bioethical regulations, intellectual property rights, and responsible use of biological information in research and society.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Baxevanis, A. D., Bader, G. D., & Wishart, D. S. (Eds.). (2020). Bioinformatics. John Wiley & Sons.
- Gautham, N. (2006). Bioinformatics: databases and algorithms. Alpha Science Int'l Ltd.
- Ham, B. M. (2011). Proteomics of biological systems: protein phosphorylation using mass
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First Year	CORE COURSE – XI (CC)	Credit : 5
Semester - IV	BIOINSTRUMENTATION	Sub Code: P25BTCC42

(Theory)

LEARNING OBJECTIVES

- The student shoul have understood the analytical techniques in the field of Biotechnology
- To make the students to understand the basic principles of Bioanalytical instruments

UNIT	CONTENTS	HOURS
I	Microscopic Techniques: Principles and Applications:Compound, Light, Stereo, Phase Contrast, FluorescentMicroscopy, Scanning and Transmission Electron Microscopy.	10
II	Centrifugation: Principle and Applications of various types of centrifugation, Density Gradient Centrifugation. Chromatography Techniques: Principle and Application of Paper Chromatography, TLC, Ion Exchange Chromatography, Affinity Chromatography, HPLC.	15
III	Electrophoretic Techniques: Principle and Application of Agarose Gel Electrophoresis, 2D-gelElectrophoresis,SDS PAGE, Iso-electric Focusing, ELISA, RIA, Southern, Northern and Western Blotting. PCR and RT-PCR, Microarray (DNA, Proteins)	15
IV	Spectroscopic Techniques: Theory and Application of UV and Visible Spectroscopy, Fluorescence Spectroscopy, Mass Spectroscopy, IR Spectroscopy NMR.	10
V	Radio-isotopic Techniques: Introduction to Radioisotopes, Uses and their Biological Applications, GM Counter, Solid and Liquid Scintillation Counter, Autoradiography, RIA, Health effects of Radiations.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles, scope, and applications of bioinstrumentation in biological and medical sciences	K1
CO2	Describe the working principles and components of commonly used bioinstrumentation devices and sensors	K2
CO3	Explain the operation and applications of laboratory instruments such as spectrophotometers, centrifuges, microscopes, and electrophoresis systems.	K3
CO4	Analyze the performance, calibration, .and selection of bioinstrumentation techniques for biological measurements and diagnostics	K3
CO5	Demonstrate the use, maintenance, and safety procedures of bioinstrumentation equipment in laboratory and clinical settings.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Introduction Practical Biochemistry – S. K. Sawhney and Randhir Singh.
- Narosa Publishing House
- Principles of Applied Biomedical Instrumentation- Cedder A and L. E. Balsar,John
- Wiley and Sons.
- Modern Experimental Biochemistry 2nd Edition- Boyer, Rodney F.Benjamin and Cummins

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Second Year	SKILL ORIENTED COURSE – I (SOC)	Credit : 5
Semester -IV	BIOENTREPRENEURSHIP, BIOPATENTING, BIOSAFETY	Sub Code: P25BTSC1

(Theory)

LEARNING OBJECTIVES

- To develop entrepreneurial skills in biotechnology and life sciences
- To ensure safe handling of biological materials and microorganisms
- To understand the importance of intellectual property rights in biotechnology

UNIT	CONTENTS	HOURS
I	Introduction to Bioethics Need for bioethics in social and cultural issues. Bioethics & GMO's Issues and concerns pertaining to Genetically modified foods & food crops, Bioethics in prenatal diagnosis, gene therapy, Organ transplantation, Xenotransplantation, regulations on field experiments and release of GMO's labelling of GM foods	10
II	Clinical trials – preclinical studies and clinical studies regulations. Bioethics & Cloning permissions and procedures in Animal cloning. Bioethics in the use of animals in research, human volunteers for Clinical research, Studies on ethnic races. Ethics in patient care, Informed consent.	15
III	Biosafety – Biological risk assessment. Biological agents and Hazard groups. Criteria in biological risk assessment. Guidelines for categorization of Genetically modified plants for field test. Regulation, national and international guidelines of Biosafety, r DNA guidelines, Regulatory requirements for drugs and biologics GLP. Biosafety levels. Safety equipments and Biological Safety cabinets.	15
IV	IPR: Introduction to Intellectual Property rights, Patenting – Factors for patentability – Novelty, Non-obviousness, Marketability. Procedures for registration of patents. Copyright works, ownership, transfer and duration of copyrights. Renewal and Termination of Copyrights. Infringements. Right of Goodwill, Passing Off. Trademarks – Introduction to Trademarks. Need for Protection of Trademarks. Classification of Trademarks Law. Procedural Requirements of protection of Trademarks	10
V	Geographical Indication- Indication of source and Geographical Indication. Procedure for Registration, Duration of Protection and Renewal. Infringement, Penalities and Remedies. Layout – Designs of Integrated Circuits: Conditions and procedure for Registration. During and effect of Registration Protection of Plant breeders' rights in India. Protection of traditional knowledge, Bioprospecting and biopiracy. India's new IP Policy (2016), Govt of India's steps to promote IPR. Career opportunities in IP. Entrepreneurship: Definition and importance, Characteristics and functions of an entrepreneur.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and principles of Bioentrepreneurship, Biopatenting, and Biosafety in biotechnology.	K1
CO2	Describe the principles of entrepreneurship, intellectual property rights, patent systems, and biosafety regulations.	K2
CO3	Apply concepts of business planning, patent filing procedures, and biosafety practices in biotechnology-related activities.	K3
CO4	Analyze the role of innovation, commercialization, and legal frameworks in biotechnology industries and research.	K3
CO5	Demonstrate knowledge and professional skills related to entrepreneurship development, intellectual property protection, and biosafety compliance.	K4
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

- Bioethics & Biosafety by Sateesh MK, IK International publications, 2008
- USPTO Web patent Database at: www.uspto.gov/patft
- Government of India's patents Website: patinfo.nic.in
- Indian patent Law: Legal and Business Implications" by Ajit Parulekar, Sarita D'Souza Macmillan india publication, 2006
- Agriculture and Intellectual property rights", edited by: Santaniello V., Evenson, R.E.,Zilberman, D. And Carlson, G.A.University press publication, 2003
- Research paper and reports provided from time to time.
- Elad Harison (2008). Intellectual property rights, Innovation and Software Technologies, Edward Elgar Publishing Limited, UK.

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MSC BIOTECH

SEM	Course	course	Subject	course code
I	Core Course-I (CC)	CC1	Cell and Molecular Biology	P25BTCC11
	Core Course-II(CC)	CC2	Biochemistry	P25BTCC12
	Core Course-III(CC)	CC3	Microbiology	P25BTCC13
	Core Practical – I (CP)	CP1	Cell and Molecular Biology, Biochemistry, Microbiology(P)	P25BTCP1
	Elective Course-I(EC)	EC1	A. Enzymology	P25BTEC1A
	Elective Course-I(EC)	EC1	B. Developmental Biology	P25BTEC1B
II	Core Course-IV(CC)	CC4	Immunology	P25BTCC21
	Core Course-V(CC)	CC5	Genetic Engineering	P25BTCC22
	Core Course -VI(CC)	CC6	Plant and Animal Biotechnology	P25BTCC23
	Core Practical – II (CP)	CP2	Immunology, Genetic Engineering, Plant and Animal Biotechnology (P)	P25BTCP2
	Elective Course-II(EC)	EC2	A. Medical Biotechnology	P25BTEC2A
	Elective Course-II(EC)	EC2	B. Nano Biotechnology	P25BTEC2B
	Non-Major Elective Course-I	NME1	A. Molecular Biotechnology	P25BTNM1A
		NME1	B. Healthcare Technology	P25BTNM1B
III	Core course-VII(CC)	CC7	Research Methodology	P25BTCC31
	Core course-VIII(CC)	CC8	Bioprocess Technology	P25BTCC32
	Core course -IX(CC)	CC9	Environmental Biotechnology	P25BTCC33
	Core Practical – III (CP)	CP3	Bioprocess Technology, Environmental Biotechnology(P)	P25BTCP3
	Elective Course III(EC)	EC3	A. Food Biotechnology	P25BTEC3A
	Elective Course III(EC)	EC3	B. Microbial biotechnology	P25BTEC3B
	Non-Major Elective Course- II	NME2	A. Agricultural Biotechnology	P25BTNM2A
	Non-Major Elective Course- II	NME2	B. Marine Biotechnology	P25BTNM2B
	Internship	INIS	Internship	P25BTIS
IV	Core Course-X(CC)	CC10	Bioinformatics and Bioethics	P25BTCC41
	Core Course-XI(CC)	CC11	Bioinstrumentation	P25BTCC42
	Skill Oriented Course I (SOC)	SOC1	Bioenterpernership, Biopatenting, Biosafety	P25BTSC1
	Project	PW	Project	P25BTPW